

NANYANG TECHNOLOGICAL UNIVERSITY

MIDTERM I

MH2100 – Calculus III

September 2021

TIME ALLOWED: 90 minutes

Name:

Matric. no.:

Tutor group:

INSTRUCTIONS TO CANDIDATES

1. **DO NOT TURN OVER PAPER UNTIL INSTRUCTED.**
2. This midterm paper contains **THREE (3)** questions.
3. Answer **ALL** questions. The marks for each question are indicated at the beginning of each question.
4. Candidates can write anywhere on this midterm paper.
5. **ONE** two-sided A4 piece of paper with handwritten notes is permitted.

QUESTION 1.**(17 marks)**

- (i) [3 marks] Find parametric equations for the line that is tangent to the curve parametrised by

$$\mathbf{r}(t) = \begin{pmatrix} 5 \sin t \\ t^3 \\ 4e^{3t} \end{pmatrix},$$

at the point $(0, 0, 4)$.

- (ii) [3 marks] The cartesian equation $y = ax + b$ describes the line tangent to the curve

$$\begin{cases} x(t) = \sin(2t) \\ y(t) = t \cos t \end{cases}$$

at the point defined by $t = \pi/2$. Find the values of a and b .

- (iii) [3 marks] At the point defined by $t = a$ the tangent vector of the curve parametrised by

$$\mathbf{r}(t) = \begin{pmatrix} e^{\sin t} \tan t \\ t^2 - 10t \\ 23 \end{pmatrix},$$

is orthogonal to the vector $\langle 0, 1, 2 \rangle$. Find the value of a .

- (iv) [3 marks] Find the directional derivative of the function

$$f(x, y, z) = x^3 + y^2 - 4z^2 + z \ln x$$

in the direction of the vector $\langle 1, 4, 1 \rangle$.

- (v) [5 marks] In which directions is the directional derivative of

$$f(x, y) = x^2y - y^8 + 2x$$

at $(6, 1)$ equal to zero? Write the unit vector for each direction.

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QUESTION 2.**(16 marks)**

- (i) [8 marks] Evaluate each of the following limits or show that the limit does not exist. Justify your answers.

(a)

$$\lim_{(x,y) \rightarrow (5,0)} \frac{5x^2y + 10}{\sqrt{x^2 + y^2}}$$

(b)

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3y}{\sqrt{x^2 + y^2}}$$

(c)

$$\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2 + x^2y + xy^3 + 3y^2}{x^2 + y^2}$$

- (ii) [8 marks] Define

$$f(x, y) = \begin{cases} \frac{x^3y+y^3}{x^6+y^2}; & \text{if } x^2 + y^2 \neq 0 \\ 0; & \text{if } x^2 + y^2 = 0. \end{cases}$$

- (a) Is f continuous at $(0, 0)$? Justify your answer.
(b) Find $f_x(0, 0)$ and $f_y(0, 0)$.
(c) Is f differentiable at $(0, 0)$? Justify your answer.

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QUESTION 3.**(17 marks)**

- (i) [3 marks] Find the directional derivative of the function

$$f(x, y) = \begin{cases} \frac{xy+y^2}{x^2+y^2}; & \text{if } x^2 + y^2 \neq 0 \\ 0; & \text{if } x^2 + y^2 = 0, \end{cases}$$

in the direction of the vector $\langle 3, 4 \rangle$ at the point $(0, 0)$.

- (ii) [3 marks] Find an equation for the tangent plane to the surface given by the equation $z = 5x^2 - 10y^2$ at the point $(2, 1, 10)$.

- (iii) [3 marks] Use the Chain Rule to find $\frac{\partial z}{\partial u}$ when $u = 0$ and $v = 2$ for

$$z = \sin(xy) + \sin(y), \quad x = 2u^2 + 3v^2, \quad y = 2uv.$$

- (iv) [3 marks] Find $\frac{\partial z}{\partial y}$ at the point $(\pi, 2\pi, \pi/2)$ where z is defined implicitly by

$$3 \cos(x + 2y) + 3 \cos(x + 2z) + \cos(2y + z) = 0.$$

- (v) [5 marks] Let

$$f(x, y) = 27x^3 - 81x + 2y^2 + 8y - 17.$$

Find and classify all critical points of f . (For each critical point, determine whether or not it is a local max/min or a saddle point.)

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